

2D Radar-to-Radar Extrinsic Calibration from Ego-Velocity Estimates:

a Python re-implementation and reproduction of Cheng, Wise & Kelly (IEEE AIM 2023)

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re-implementation, simulation studies and Endeavour re-run, September 2026

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Abstract

Two 2D millimetre-wave radars rigidly mounted on a vehicle each measure their own ego-velocity from the Doppler range-rates of static scatterers. Cheng, Wise and Kelly [1] showed that the yaw between the radars and the *direction* of the translation between them can be recovered from these ego-velocity streams alone: no retroreflectors, no overlapping fields of view, no third sensor. This document records a from-scratch Python re-implementation of the method (RANSAC ego-velocity front end, batch Levenberg–Marquardt solver with Schur elimination, closed-form initialisation, observability diagnostics), the reproduction of the paper’s simulation studies (16 noise/duration settings $\times$ 100 trials), a scatterer-level end-to-end simulation, and a re-run on the Endeavour shuttle-bus dataset. On Endeavour the re-run reproduces the paper’s Table I to within 0.01 rad for two of the three radar pairs available in the archived bags, and lowers the held-out velocity-error metric of the paper’s Table III below both the dataset’s and the paper’s parameters on 11 of 13 run–pair combinations. Two findings not in the paper are documented: the July-2022 “updated” bags carry a mirrored azimuth and a 7% Doppler speed bias on the side radars, under which the batch problem prefers a degenerate solution; and the near-lateral pair Near5–Near3 is only weakly identifiable, with the translation axis recovered 23° from the dataset value.

Keywords: extrinsic calibration, mm-wave radar, ego-velocity, Doppler, batch estimation, identifiability, Endeavour dataset.

1 Overview

The original work [1] was written in MATLAB (batch solver, EKF variant, Endeavour tools); the archived code is not public. Everything here is a clean Python re-implementation in `experiments/r2r.py` (model, solver, initialisation, RANSAC, scatterer simulator), `sim_sweep.py` (paper Section V-A plus degeneracy and scale studies), `sim_pipeline.py` (raw returns $\rightarrow$ ego-velocity $\rightarrow$ calibration), `endeavour.py` (bag reader, front end, calibration, held-out evaluation), `fig_endeavour.py` and `make_tables.py` (figures and the tables in this document), `make_video.py`. Every figure in `figures/` is regenerated from scratch by these scripts with fixed seeds.

2 Method

2.1 Notation and conventions

$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ is the active planar rotation. Radars a and b share a horizontal sensing plane. $\mathbf{v}_a^j \in \mathbb{R}^2$ is the ego-velocity of radar a at time j in its own frame, ω^j the yaw rate of the rigid body, $\mathbf{t}_{ba}^a$ the translation from a to b in frame a , and $\mathbf{R}_{ba} = \mathbf{R}(\theta_{ba})$ takes frame- a coordinates to frame- b coordinates. The archived MATLAB code and the paper use the passive matrix `theta2dcm`(θ) = $\mathbf{R}(-\theta)$, so every θ_{ba} reported by the paper equals $-\theta_{ba}$ of this implementation; θ_t is identical. All Endeavour numbers below are quoted in the paper’s convention.

2.2 Ego-velocity from one radar scan

A scan returns N targets with range r_i , azimuth θ_i (from the x axis) and range-rate $\dot{r}_i$. For a static target, $\dot{r}_i = -[\cos \theta_i \ \sin \theta_i] \mathbf{h}$, so $\mathbf{h}$ follows from linear least squares (paper eq. 3–7) with covariance $\Sigma = \frac{\epsilon^\top \epsilon}{N-2} (A^\top A)^{-1}$. Moving targets and multipath are removed by a two-point RANSAC (200 hypotheses drawn at once, pairs with $|\sin(\theta_i - \theta_j)| < 0.1$ rejected, inlier threshold τ , at least $\max(8, 0.3N)$ inliers), followed by a final least-squares refit on the inliers. The paper quotes $\tau = 0.025$ m/s; on the archived ARS430 bags the inlier residual spread is larger (§3.6) and $\tau = 0.1$ m/s is used, which is also the value used for the scatterer simulation.

2.3 Measurement model and batch problem

Because $\omega \times \mathbf{t}$ in the plane is $\omega |\mathbf{t}| \mathbf{u}_\perp$ with $\mathbf{u} = [\cos \theta_t, \sin \theta_t]^\top$ and $\mathbf{u}_\perp = [-\sin \theta_t, \cos \theta_t]^\top$, the two ego-velocity measurements are (paper eqs. 8–9)

$$\mathbf{h}_a^j = \mathbf{v}_a^j + \mathbf{n}_a^j, \quad \mathbf{h}_b^j = \mathbf{R}(\theta_{ba})(\omega_\gamma^j \mathbf{u}_\perp + \mathbf{v}_a^j) + \mathbf{n}_b^j, \quad \omega_\gamma^j \equiv \omega^j |\mathbf{t}_{ba}|. \quad (1)$$

Scaling ω_γ^j by γ and $\mathbf{t}_{ba}$ by $1/\gamma$ leaves (1) unchanged (paper §III-E), so only the *unscaled* yaw rate ω_γ^j and the translation *axis* $\theta_t \in [0, \pi)$ are identifiable; the axis has period π because $\theta_t \rightarrow \theta_t + \pi$ with $\omega_\gamma \rightarrow -\omega_\gamma$ is the same measurement. The batch estimate minimises

$$J(\mathbf{x}) = \sum_{j=1}^M (\mathbf{h}_a^j - \mathbf{v}_a^j)^\top \Sigma_a^{j-1} (\mathbf{h}_a^j - \mathbf{v}_a^j) + \mathbf{e}_b^{j\top} \Sigma_b^{j-1} \mathbf{e}_b^j, \quad \mathbf{e}_b^j = \mathbf{h}_b^j - \mathbf{R}(\theta_{ba})(\omega_\gamma^j \mathbf{u}_\perp + \mathbf{v}_a^j), \quad (2)$$

over $\mathbf{x} = [\mathbf{v}_a^1, \omega_\gamma^1, \dots, \mathbf{v}_a^M, \omega_\gamma^M, \theta_t, \theta_{ba}]$ ($3M + 2$ unknowns, $4M$ residuals).

2.4 Solver

The Jacobians are analytic: $\partial \mathbf{e}_b / \partial \mathbf{v}_a = -\mathbf{R}$, $\partial \mathbf{e}_b / \partial \omega_\gamma = -\mathbf{R} \mathbf{u}_\perp$, $\partial \mathbf{e}_b / \partial \theta_t = \omega_\gamma \mathbf{R} \mathbf{u}$, $\partial \mathbf{e}_b / \partial \theta_{ba} = -\mathbf{R}'(\theta_{ba})(\omega_\gamma \mathbf{u}_\perp + \mathbf{v}_a)$. Levenberg–Marquardt normal equations are solved by eliminating the M independent 3×3 per-time blocks (Schur complement onto the two calibration parameters), so one iteration costs $\mathcal{O}(M)$ and a 120-s problem ($M = 1681$) solves in 60 ms. After each step θ_t is wrapped to $[0, \pi)$ (flipping the sign of all ω_γ^j if it crossed π) and θ_{ba} to $(-\pi, \pi]$. The covariance of (θ_t, θ_{ba}) is the inverse of the Schur complement of the undamped Gauss–Newton information, and its smallest eigenvalue is the identifiability diagnostic used in §3.3.

2.5 Initialisation (paper §III-F)

θ_{ba}^0 is the circular median of $\text{atan2}(\mathbf{h}_a \times \mathbf{h}_b, \mathbf{h}_a \cdot \mathbf{h}_b)$ over the pairs whose speeds agree to within 0.05 m/s (the rotation term is then negligible); $\mathbf{b}^j = \mathbf{R}^\top \mathbf{h}_b^j - \mathbf{h}_a^j \approx \omega_\gamma^j \mathbf{u}_\perp$ gives θ_t^0 as the $|\mathbf{b}^j|$ -weighted doubled-angle mean of $\angle \mathbf{b}^j - \pi/2$ modulo π , and $\omega_\gamma^{j,0} = \mathbf{b}^j \cdot \mathbf{u}_\perp$. With (θ_t, θ_{ba}) frozen the problem is linear in $(\mathbf{v}_a^j, \omega_\gamma^j)$ and is solved in closed form per time step before the LM iterations start.

2.6 Identifiability

The paper proves local weak observability with the Hermann–Krener rank condition and derives $\det \mathbf{O} \propto \alpha_\gamma (\mathbf{h}_a \times \mathbf{u})$ (paper eq. 26): the yaw rate must change ($\alpha_\gamma \neq 0$), the platform must move, and the ego-motion must not be aligned with the translation axis. The re-implementation checks this numerically: Fig. 5 plots $\alpha_\gamma (\mathbf{h}_a \times \mathbf{u})$ along the nominal trajectory, and §3.3 sweeps both degenerate directions and reports the smallest eigenvalue of the Schur information, which goes to zero exactly when the motion is aligned with the axis.

2.7 Metric scale from a coarse yaw-rate source (paper §V-B)

If any sensor provides a yaw rate $\tilde{\omega}^j$, then $|\mathbf{t}_{ba}| = |\omega_\gamma^j|/|\omega^j|$; the re-implementation uses the weighted least-squares estimate $|\mathbf{t}_{ba}| = \sum_j |\omega_\gamma^j| |\tilde{\omega}^j| / \sum_j \tilde{\omega}^{j,2}$ over the times with $|\tilde{\omega}^j| > 0.1$ rad/s.

3 Experiments

3.1 Simulation sweep (paper §V-A, Fig. 3)

The nominal trajectory of the archived `gen_data_2.m` is used: $\mathbf{v}_a(t) = [\sin 2t + 5, 0.25 \cos 2t + 1]$ m/s, $\omega(t) = \max(0.3 \sin(t/2), 0)$ rad/s (Fig. 2), 14 Hz, $\mathbf{t}_{ba} = [-5, 0]$ m ($\theta_t = 0$), $\theta_{ba} = \pi/2$. Isotropic noise $\sigma_r \in \{0.05, 0.10, 0.15, 0.20\}$ m/s is added to both ego-velocities, durations are $\{15, 30, 60, 120\}$ s, 100 seeded trials each (1600 solves, 141 s total).

Table 1 and Fig. 1 reproduce the paper’s statement that the estimates are within 2° and 3° “for most noise levels and durations”: every trial with $\sigma_r \leq 0.1$ m/s and $T \geq 30$ s is inside the envelope, and the medians are below 0.7° everywhere. The failures are all of one kind: for the shortest, noisiest settings (15 s at $\sigma_r \geq 0.1$ m/s, 30 s at $\sigma_r = 0.2$ m/s) between 2 % and 24 % of trials converge to a wrong local minimum about 20° away in both parameters, because the closed-form initialisation is computed from too few informative (turning) samples. The reported 1σ values track the empirical spread (Fig. 1).

3.2 Fused ego-velocity (paper Fig. 4)

At $\sigma_r = 0.2$ m/s and 120 s the batch solution lowers the median ego-velocity error of both radars from 0.231 m/s and 0.244 m/s to 0.192 m/s and 0.206 m/s (Fig. 3); averaged over 100 trials the improvement is 0.036 ± 0.003 m/s for radar *a* and 0.035 ± 0.004 m/s for radar *b*. The paper reports 4 cm/s; the difference is within the trial-to-trial spread of the raw medians.

3.3 Degenerate motions

Fig. 4 sweeps the two degeneracies of §2.6 at $\sigma_r = 0.1$ m/s, 60 s, 20 trials each. Left: the yaw-rate oscillation amplitude is reduced to zero (constant $\omega = 0.15$ rad/s, so $\alpha_\gamma = 0$): the errors

Table 1: Simulation sweep: absolute errors of the translation axis θ_t and rotation θ_{ba} over 100 trials (median / 95th percentile / maximum, degrees), the median reported 1σ of each parameter, and the fraction of trials inside the paper’s $2^\circ/3^\circ$ envelope.

σ_r [m/s]	T [s]	$ \Delta\theta_t $ [deg]			$ \Delta\theta_{ba} $ [deg]			σ_{θ_t} [deg]	$\sigma_{\theta_{ba}}$ [deg]	in $2^\circ/3^\circ$
		med	p95	max	med	p95	max			
0.05	15	0.19	0.61	1.1	0.24	0.88	1.1	0.36	0.36	100%
0.05	30	0.17	0.46	0.6	0.18	0.50	0.8	0.24	0.26	100%
0.05	60	0.10	0.35	0.4	0.12	0.35	0.5	0.18	0.18	100%
0.05	120	0.08	0.27	0.4	0.09	0.26	0.4	0.13	0.12	100%
0.10	15	0.54	1.67	20.6	0.39	1.49	16.8	0.71	0.72	98%
0.10	30	0.37	0.96	1.1	0.37	0.93	1.2	0.48	0.52	100%
0.10	60	0.24	0.82	1.0	0.27	0.74	1.1	0.36	0.35	100%
0.10	120	0.16	0.45	1.0	0.19	0.61	0.7	0.26	0.25	100%
0.15	15	0.70	2.21	20.8	0.71	2.43	15.6	1.05	1.09	89%
0.15	30	0.40	1.59	19.8	0.51	1.72	14.7	0.71	0.78	96%
0.15	60	0.42	1.03	1.6	0.49	1.29	1.6	0.54	0.53	100%
0.15	120	0.29	0.77	1.2	0.26	0.87	1.0	0.38	0.37	100%
0.20	15	1.05	19.45	21.8	1.08	13.36	18.9	1.36	1.48	76%
0.20	30	0.67	3.56	21.7	0.70	4.43	18.8	0.92	1.03	88%
0.20	60	0.63	1.50	2.4	0.50	1.28	1.6	0.70	0.68	99%
0.20	120	0.38	1.02	1.2	0.28	1.07	1.6	0.49	0.49	100%

grow from $0.11^\circ/0.17^\circ$ at amplitude 0.2rad/s to $0.42^\circ/0.98^\circ$ at zero amplitude and the smallest information eigenvalue drops 15-fold, but the problem does not become singular because the ego-velocity direction still varies (the $\mathbf{h}_a \times \mathbf{u}$ factor). At amplitude 0.3rad/s the yaw rate changes sign during the run and the closed-form initialisation, which assumes a dominant sign, fails (20° median error): a limitation of the initialiser, not of the batch problem, since starting from the truth converges. Right: the ego-motion direction is held at an angle ψ from the translation axis. At $\psi = 0$ the smallest eigenvalue is numerically zero (-10^{-9}), i.e. the pair of parameters is unobservable exactly as the determinant predicts; at $\psi \geq 5^\circ$ the errors are back at the nominal 0.35° level.

3.4 Metric scale from a coarse gyro (paper Table IV)

With a simulated yaw-rate source of 0.02rad/s noise, the translation magnitude over 100 trials (120s, $\sigma_r = 0.1\text{m/s}$) is $4.96 \pm 0.09\text{m}$ against 5m (Fig. 6).

3.5 Scatterer-level pipeline

To exercise the front end and the solver together, `sim_pipeline.py` drives a vehicle for 60s through 900 random point scatterers (Fig. 7); each radar (120° field of view, 45m range, 0.05m/s range-rate and 1° azimuth noise, 12% moving-target outliers) produces a scan at 14Hz, from which RANSAC estimates the ego-velocity. Radar b sits at $[-3, 1.2]\text{m}$ (3.23m) and 95° from radar a . 68% of the scans of radar a and 90% of radar b yield a valid ego-velocity (the rest fail the inlier count); the ego-velocity RMS error is 0.031m/s and 0.026m/s with medians of 0.021 and 0.021m/s . The calibration from the valid pairs, weighted with the per-scan covariances, gives $\theta_t = 158.06^\circ$ (truth 158.20) and $\theta_{ba} = 95.11^\circ$ (truth 95.00), reported 1σ 0.05° and 0.02° , in 4 LM iterations from a closed-form initialisation that was already within $0.18^\circ / 0.13^\circ$ (Fig. 8). With isotropic weights the errors are 0.14° and 0.10° ; the fused velocities have median errors of 0.019 and 0.016m/s .

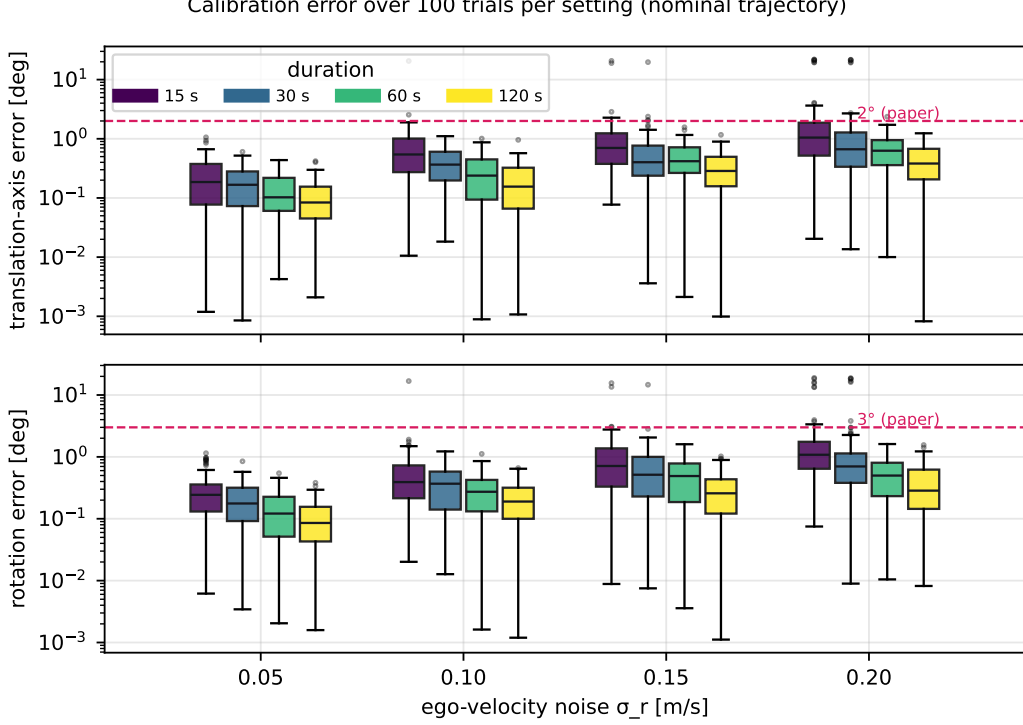


Figure 1: Error distributions per noise level (x axis) and duration (colour), 100 trials each; dashed lines are the paper’s 2° and 3° envelopes. Log scale; the isolated $\sim 20^\circ$ points are initialisation failures at the shortest and noisiest settings.

3.6 Endeavour dataset (paper §V-B)

Data. Five Continental ARS430 radars on the Xi’an Jiaotong “Endeavour” shuttle bus [7], 13 Hz, six runs (East/Mid/West $\times$ 2) of 4–12 minutes. Two archived conversions of the raw cluster lists to PointCloud2 exist. In the July-2022 “updated” bags the point coordinates satisfy $\text{atan2}(x, y) = -\phi + 0.057$ rad where ϕ is the sensor’s own azimuth field, i.e. the azimuth is mirrored and offset; the ego-velocities computed from them over-estimate the odometry speed by 6.7–6.9 % on the four side radars and 2.6 % on the front radar, and with such a between-radar scale mismatch the batch problem (which can absorb any residual along $\mathbf{u}_\perp$ through the free ω_γ^j) converges to a degenerate solution with $\theta_t \approx 93^\circ$ for every pair and θ_{ba} 25°–30° off. In the September-2022 “corrected-orientation” bags $\text{atan2}(x, y) = \phi$ exactly, the speed ratio is 0.995–0.999 on all radars, and every ego-motion direction is where the mounting says it should be ($+y$ for the front radar Near5, $\pm x$ for the side radars). All results below use the corrected bags. They contain radars Near1, 2, 3, 5 only (Near4 is absent from all six, Near1 from West 1/2), so the pairs Near5–Near {1,2,3} are calibrated on East 2 and evaluated on the other runs.

Front end. Points with range < 1 m or $|r| > 10$ m/s are dropped; RANSAC with $\tau = 0.1$ m/s then keeps a median of 66–92 inliers out of 72–110 returns per scan, and 99.8 % of the scans yield an estimate (with the paper’s $\tau = 0.025$ m/s only 5–30 % do, because the range-rate spread of this radar is 0.05 m/s-class). Fig. 9 compares $|\mathbf{h}|$ with the odometry speed.

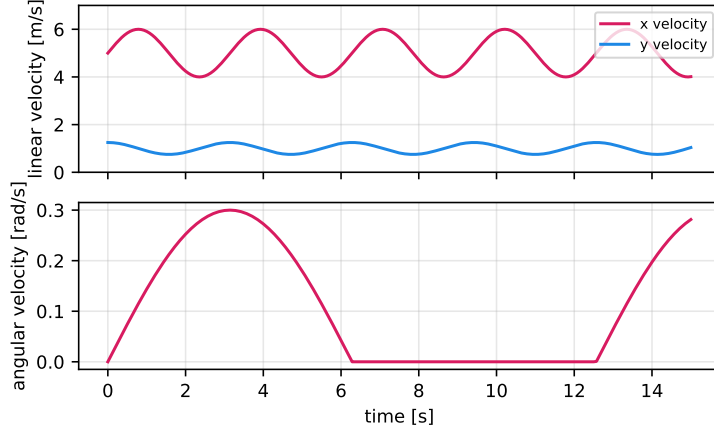


Figure 2: The nominal trajectory (paper Fig. 2): ego-velocity of radar a and yaw rate over the 15-s period.

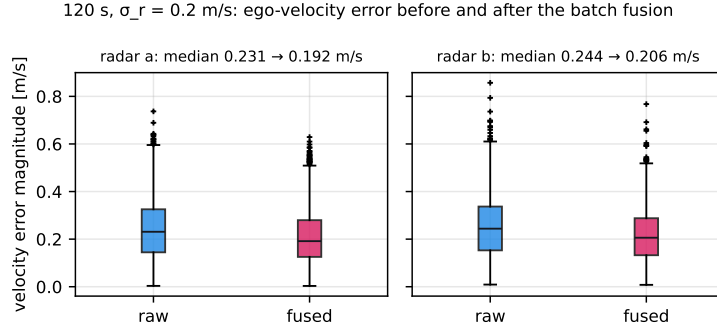


Figure 3: Raw versus fused ego-velocity error magnitude, one 120-s trial at $\sigma_r = 0.2$ m/s.

Calibration (paper Table I). Radar b is interpolated onto radar a ’s timestamps, pairs further than 0.2s from a sample or slower than 0.05 m/s are dropped ($M \approx 6470$ per pair), and the per-scan covariances (floored at 10^{-4}) weight the residuals. Table 2 lists the result next to the dataset’s own extrinsics and the paper’s estimates, both quoted in the paper’s convention. Both initialisations (automatic and at the dataset values) converge to the same solution for every pair.

For Near 5–Near 1 and Near 5–Near 2 the re-run lands within 0.015 rad of the paper’s own 2023 estimates (1.754/−1.583 vs. 1.74/−1.58; 1.409/1.615 vs. 1.41/1.61) and within 2.5° of the dataset extrinsics, which is the paper’s “within 3°” claim. For Near 5–Near 3 the rotation agrees to 0.1° but the translation axis comes out at 3.13 rad instead of 2.73 (dataset) / 2.79 (paper): the baseline of this pair is only 0.83 m, so the lever-arm term $\omega|\mathbf{t}|\mathbf{u}_\perp$ is at most 0.25 m/s and the axis is set by a signal comparable to the ego-velocity noise; the reported 1σ of 0.06° is therefore over-confident, and the held-out evaluation (below) confirms that the dataset’s axis explains the data slightly better for this pair.

Held-out evaluation (paper Table III). On each of the other runs, and for each parameter set, ω_γ^j is chosen per time to minimise $|\mathbf{e}_b^j|$ with $\mathbf{v}_a^j = \mathbf{h}_a^j$, and the mean $|\mathbf{e}_b^j|$ is reported (Table 3, Fig. 11). The re-run parameters give the lowest error on every run for Near 5–Near 1 and Near 5–Near 2 (0.009–0.018 m/s, against 0.017–0.028 m/s with the dataset values), the same ordering and

Degenerate motions: 20 trials each, $\sigma_r = 0.1$ m/s, 60 s

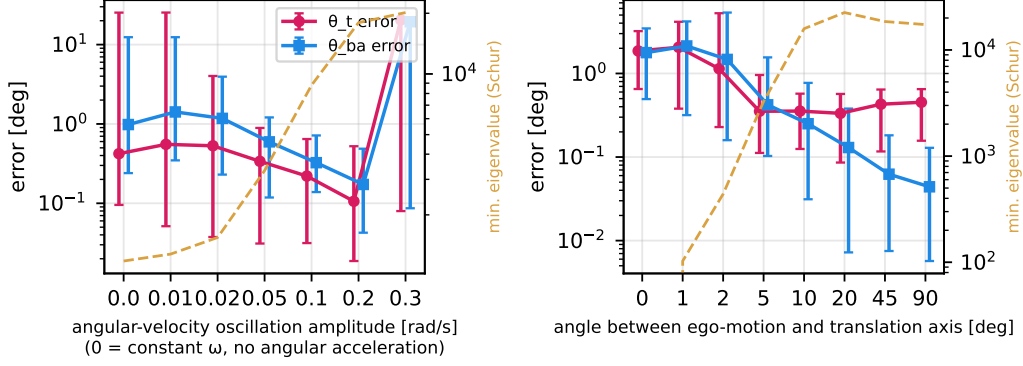


Figure 4: Degeneracy sweeps: no angular acceleration (left) and ego-motion aligned with the translation axis (right). Bars are 10–90 % ranges over 20 trials; the gold dashed line is the smallest eigenvalue of the Schur-complement information of (θ_t, θ_{ba}) .

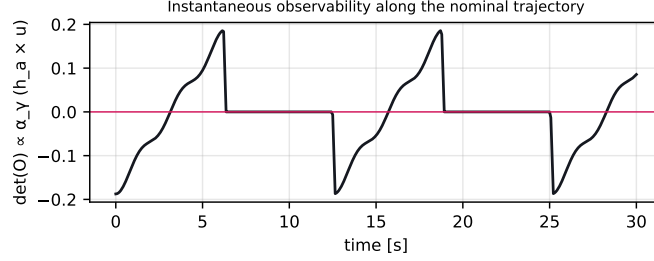


Figure 5: $\det \mathbf{O} \propto \alpha_\gamma(\mathbf{h}_a \times \mathbf{u})$ (paper eq. 26) along the nominal trajectory: it is zero whenever the yaw rate is constant, which happens half of every 12.6-s period.

magnitude as the paper’s Table III; for Near5–Near3 the dataset parameters win on four of five runs, consistent with the weak identifiability of that pair.

4 Implementation and Usage

```
python3 -m venv .venv && .venv/bin/pip install numpy scipy matplotlib rosbags
cd experiments
python sim_sweep.py           # 1600 solves, ~2.5 min: sim_sweep.json + figures
python sim_pipeline.py        # scatterer-level pipeline + video data
ENDEAVOUR_SET=corr python endeavour.py east2 east1 mid1 mid2 west1 west2 stage2
python fig_endeavour.py && python make_tables.py && (cd ../docs && make)
python make_video.py          # thumbnail + long demo video (needs ffmpeg)
```

The library is `r2r.py`: `calibrate(h_a, h_b, cov_a, cov_b)` returns (θ_t, θ_{ba}) , the fused states and covariance; `ego_velocity_ransac(az, rr)` is the front end; `RadarScanSim` the scatterer world. The bags are read with the pure-Python `rosbags` package (no ROS). Hardware path: any radar driver that publishes per-target azimuth and range-rate feeds `ego_velocity_ransac` directly; the calibration itself only needs the two ego-velocity streams and their timestamps.

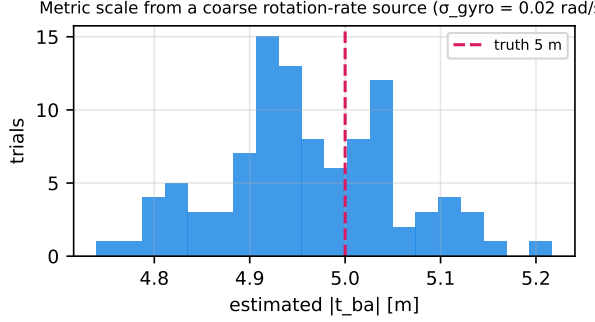


Figure 6: Recovered translation magnitude with a coarse gyro.

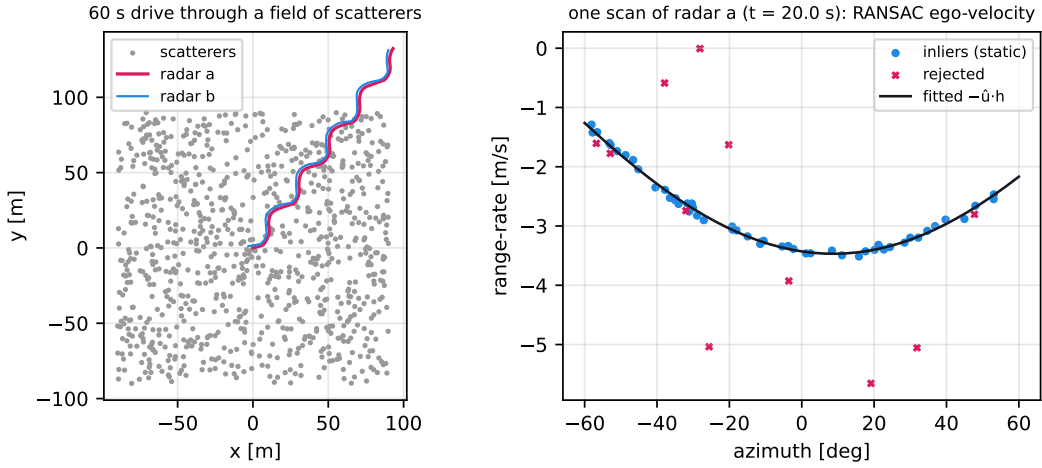


Figure 7: Scatterer simulation: trajectory of both radars (left) and one scan of radar a with the RANSAC fit of $-\hat{\mathbf{u}} \cdot \mathbf{h}$ (right).

5 Limitations

- The archived corrected bags lack Near4 entirely, so the paper’s fourth pair could not be re-run, and Near1 is missing from the West runs.
- Near5–Near3 is weakly identifiable (short baseline); the batch covariance under-reports the error there. A prior on θ_t or a longer/more excited run would be needed.
- The closed-form initialisation fails in a few percent of short, noisy simulated runs and whenever the yaw rate changes sign with small amplitude; the batch problem itself is fine from a better start.
- The Olutomilayo et al. retroreflector-map baseline of the paper’s Tables I–III was not re-implemented (it needs the lidar map and the stationary segments).
- The RANSAC threshold is a per-radar tuning constant; the ARS430 needs 0.1 m/s, not the 0.025 m/s of the paper’s text.
- Ego-velocity residual covariances from the least-squares fit are optimistic when a scan has few inliers; they are floored at $10^{-4} \text{ m}^2/\text{s}^2$.

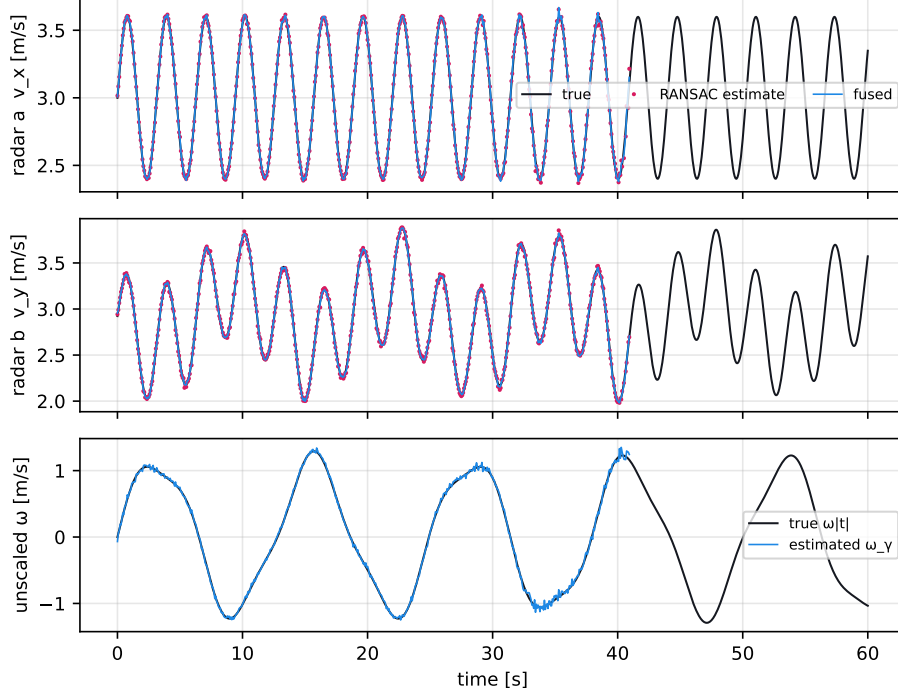


Figure 8: True, RANSAC-estimated and fused velocities, and the recovered unscaled yaw rate ω_γ (which equals $\omega|\mathbf{t}|$ with $|\mathbf{t}| = 3.23$ m).

Table 2: East 2 calibration (radar a = Near 5). Angles in radians in the paper’s convention; σ are the reported 1σ in degrees; the last two columns are the differences to the dataset values in degrees.

pair	M	dataset		paper (ours 2023)		this re-run		σ_{θ_t} [deg]	$\sigma_{\theta_{ba}}$ [deg]	$\Delta\theta_t$ [deg]	$\Delta\theta_{ba}$ [deg]
		θ_t	θ_{ba}	θ_t	θ_{ba}	θ_t	θ_{ba}				
Near 5–Near 1	6477	1.71	-1.57	1.74	-1.58	1.754	-1.583	0.05	0.017	+2.5	+0.8
Near 5–Near 2	6472	1.44	1.59	1.41	1.61	1.409	1.615	0.05	0.016	-1.9	-1.6
Near 5–Near 3	6478	2.73	-1.58	2.79	-1.58	3.134	-1.573	0.06	0.003	+23.2	-0.1

References

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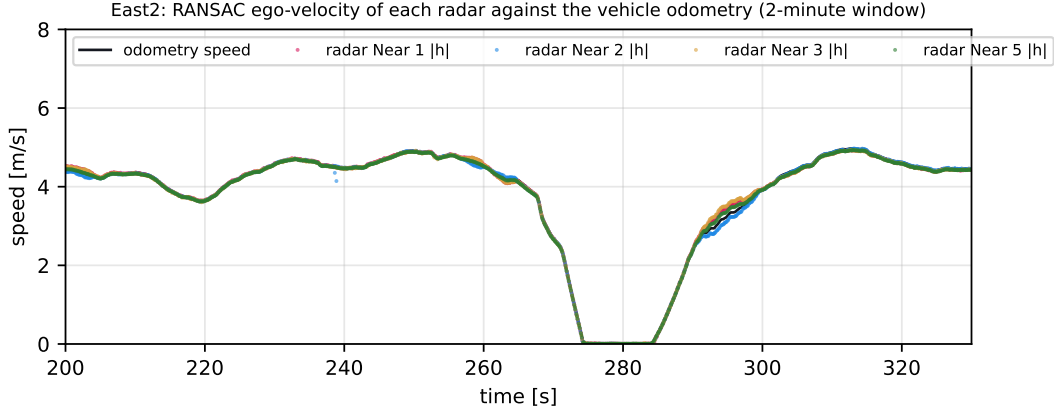


Figure 9: East 2: ego-velocity magnitudes of the four radars against the odometry speed.

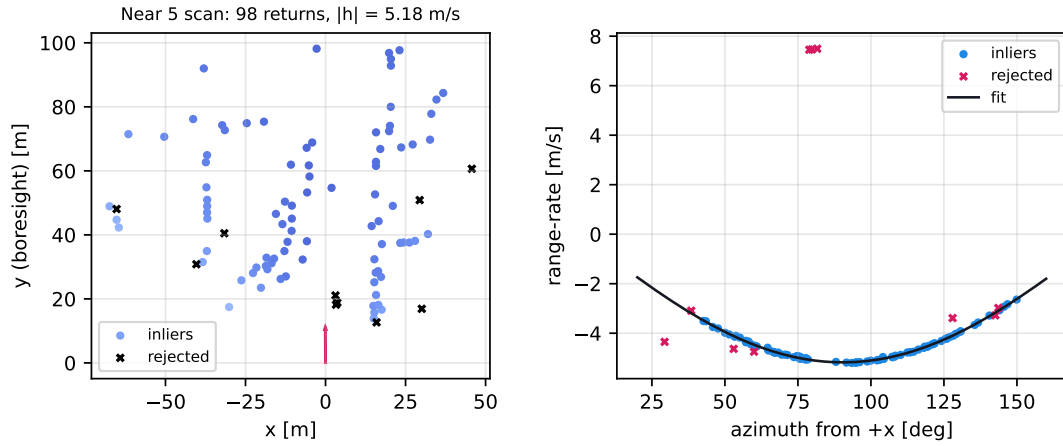


Figure 10: One real scan of the front radar with the RANSAC inliers, rejected returns and the fitted ego-velocity.

- [7] Y. Hu et al., “Endeavour Radar Dataset,” Xi’an Jiaotong University, 2021. https://github.com/gloryhry/radar_lidar_static_calibration.
- [8] E. Wise, Q. Cheng, J. Kelly, “Spatiotemporal calibration of 3D millimetre-wavelength radar-camera pairs,” *IEEE T-RO*, 2023.

Table 3: Mean velocity-error magnitude [m/s] on the held-out runs with the dataset parameters (d), the paper’s 2023 parameters (p) and this re-run (r); bold = lowest.

run	Near 5–Near 1			Near 5–Near 2			Near 5–Near 3		
	d	p	r	d	p	r	d	p	r
east1	0.0174	0.0095	0.0089	0.0207	0.0105	0.0099	0.0109	0.0233	0.0145
mid1	0.0284	0.0195	0.0179	0.0258	0.0136	0.0121	0.0182	0.0215	0.0189
mid2	0.0272	0.0188	0.0170	0.0264	0.0146	0.0133	0.0178	0.0191	0.0200
west1	–	–	–	0.0236	0.0112	0.0106	0.0127	0.0240	0.0154
west2	–	–	–	0.0248	0.0127	0.0122	0.0126	0.0242	0.0157

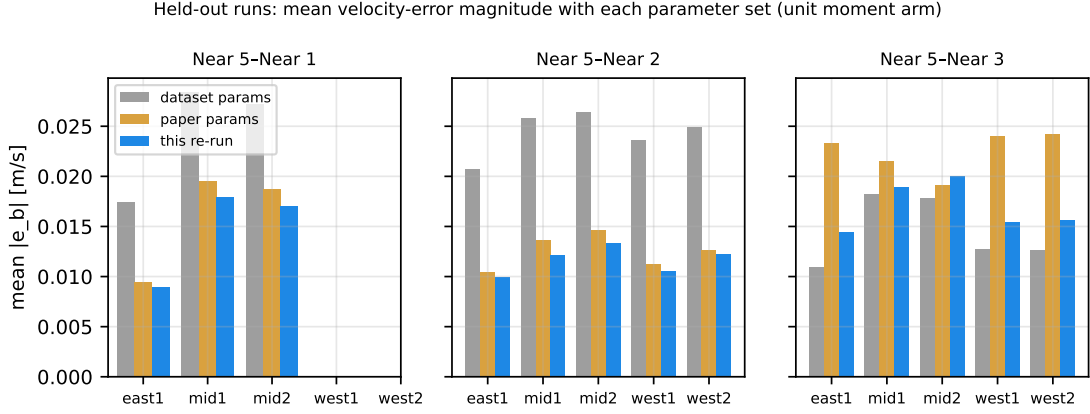


Figure 11: Held-out velocity errors per run and parameter set.

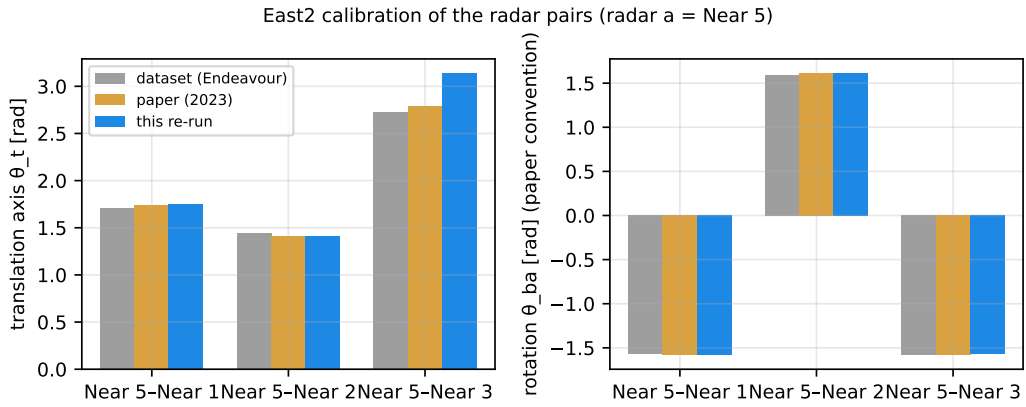


Figure 12: East 2 parameters: dataset, paper and this re-run.